MIL-T-82096B(YD)
30 May 1986
SUPERSEDING
MIL-T-82096A(YD)
01 December 1976

MILITARY SPECIFICATION

TOWER, PREFABRICATED, STEEL: FOR 4200 GALLON WATER TANK

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an elevated prefabricated sectionalized tower designed to support a 4200 gallon bolted steel, water storage tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5445

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AMSC N/A

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MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL FACILITIES ENGINEERING COMMAND

6027429 - Steel Tower, 52 FT. (ABFC Assembly 41500).

(Application for copies should be addressed to the Commanding Officer, Naval Facilities Engineering Command, 200 Stovall Street, Alexandria, VA 22332-2300)

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.
A235 - Specification for High Strength Bolts for Structural Joints, Including Suitable Nuts and Plain Hardened Washers.
D3951 - Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The tower shall be prefabricated, sectionalized, and knocked down, for erection by bolted connections in the field. The nominal dimensions, general arrangement, and design shall be as shown on Drawing 6027429. The tower shall be complete with all structural steel members, adjustable tension rods, braces, girders, joists, platform, walkway, railing, ladder, anchor bolts, base plates, and all fasteners for complete field erection by others. The bolted steel tank, water pipes, pipe fittings, valves insulation, and concrete shall not be furnished under this specification.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish a complete tower for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.3.1 Structural steel. Steel for structural or load bearing members shall conform to ASTM A36, except that structural steel having superior strength may be used.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Design. The tower shall be designed as shown on Drawing 6027429 and other requirements specified herein. The design shall provide for earthquake vertical and horizontal forces in accordance with the criteria for the seismic design standard of building and other structures.

3.6 Ladder. Ladder shall be furnished prefabricated in lengths corresponding to the tower section heights shown on the drawing. Side rails shall be not less than 2-inch steel bar 5/16-inch thick; spacing between rails to be 15 inches ± 1 inch. Rungs shall be not less than 5/8-inch round or square bars spaced at 12-inch centers. Ladder shall extend from a point 10 feet above the base up through the walkway at the top. The ladder shall be furnished without rest platforms or safety cage.

3.7 Platform. Girders and joists shall be I-beam or equivalent shapes of suitable strength and spacing to support the 4200 gallon tank of water, the walkway, and the railing.

3.8 Walkway grating. Floor grating for the walkway around the perimeter of the base of the tank shall have steel parallel bearing bars at center-to-center distance of not more than 1-3/16 inches, with right angle steel crossmembers secured to bearing bars and spaced not to exceed 4 inches. Design load-carrying capacity of the walkway shall be not less than 125 pound-force per square foot. Curved edges at perimeter shall have suitable edge bands.

3.9 Railing. The railing around the perimeter of walkway shall be prefabricated of standard weight galvanized iron pipe of galvanized steel angle, with connections and supports for field erection. The railing shall consist of top rail, intermediate rail, toe plate and posts. The railing shall be designed in accordance with OSHA Section 1926.500(f).

3.10 Fasteners. Bolts for field connection of structural members shall conform to ASTM A325. A 5 percent excess quantity of each size and type fastener shall be provided. Fasteners shall be of the type regularly used for bolted steel tower structures.

3.11 Marking. The tower components shall be marked for identification in accordance with MIL-STD-130. In addition, erection marks shall be stenciled or die cut on each component or on an attached metal tag in the case or rods

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or other narrow items. Markings shall be clearly legible and of suitable size and contrast. Identification marking shall include the nomenclature, the Federal stock number, year of manufacture, the manufacturer, and the contract number.

3.12 Erection manual. The contractor shall furnish to the government erection instructions normally furnished in commercial practice for each tower specified in the contract.

3.13 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Unless otherwise specified (see 6.2.1), color of the finish coat shall be dark olive drab. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual; nor does it commit the Government to acceptance of defective material.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one tower when a first article is required (see 3.2 and 6.3). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, and the packaging inspection of 4.5.

4.3 Examination. Each tower shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy

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6. NOTES

and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Erection test. Component parts for one complete tower shall be selected at random from the first ten or less towers furnished under the contract. The contractor shall assemble and erect the tower on a suitable level base, and shall interchange like parts on the tower structure. Boltholes shall aline properly with the aid of a rigger wrench to receive the fasteners, and assembly shall be achieved without distortion or racking of the structure. Failure of any component to fit properly shall be cause for rejection of all such components. After disassembly of the approved tower by the contractor, the reusable, repaired, or replaced components shall have touchup painting, and then may be prepared for shipment and submitted as one of the towers to be furnished under the contract.

4.5 Packaging inspection. The preservation, packing, and marking of the tower shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. The preservation shall be level A or commercial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Unpainted surfaces. Unpainted surfaces, including bolts, nuts, threaded ends of components, and surfaces exposed by disassembly shall be coated with type P-1 preservative in accordance with MIL-P-116.

5.1.1.2 Fasteners. Bolts, nuts, and washers for one complete tower shall be packaged method I or III, in close-fitting boxes conforming to PPP-B-636, class weather resistant, and closed method V in accordance with the appendix of the box specification.

5.1.1.3 Erection manuals. Erection manuals for each tower shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.2 Commercial. Material shall be packaged in accordance with ASTM D3951. When specified (see 6.2.1), flammable, combustible, or toxic packaging materials (e.g., loose-fill polystyrene, wood excelsior, shredded paper, newspaper, wax paper, etc.) shall not be used.

5.2 Packing. The packing shall be level A, B, or Commercial, as specified (see 6.2.1).

5.2.1 Level A.

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* 5.2.1.1 Boxed components. The bolts, nuts, washers, erection drawings, clips, fittings, and any other small components for one complete tower shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type. Contents shall be cushioned and braced to prevent movement. The boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be class 1, type I, finish B in accordance with QQ-S-871.

* 5.2.1.2 Bundled components. Structural frame members, ladders, platform, and railings for one complete tower shall be bundled into secured lifts not to exceed 4000 pounds. The components shall be nested, arranged, and the bundles secured girthwise with strapping in accordance with QQ-S-781, class 1, type I, finish B, or bolting in combination with suitable wood blocking or battens as required to form compact nonshifting bundles. A minimum of three (3) straps shall be provided a center strap with two end straps. Strapping shall be spaced not to exceed 36-inches on center with intermediate straps provided depending upon length of the lift. The center strap shall be spaced not to exceed one (1) from the center of the lift. The end straps shall be placed at the same distance (±1-inch) from either end of the lift and shall not be more than 18-inches from each end of the lift. End straps shall be at the same distance from the ends for all lifts of the same length. Intermediate straps shall be equally spaced for all lifts. Edge protectors of metal or pressed fiberboard shall be used where strapping bears on sharp metal edges. The strapping shall be stapled to any wood blocking or battens provided. The bundles shall have skids with a minimum clearance of 3-1/2 inches to provide handling by fork lift trucks.

* 5.2.2 Level B. The complete tower shall be packed as specified for level A, except that the boxes shall be domestic type and strapping shall be finish A.

5.2.3 Commercial. The complete tower shall be packed in accordance with ASTM D3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tower specified herein is intended for temporary or semipermanent use for an elevated water storage tank at advanced bases.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection is required for inspection and approval (see 3.2, and 6.3).
c. Color of finish paint if other than as specified (see 3.13).

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d. Level of preservation and level of packing required (see 5.1 and 5.2).
e. If flammable, combustible, or toxic packaging materials are not to be used (see 5.1.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item consisting of one complete tower or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 Subject term (key word) listing.

Elevated Water Storage Tank
Prefabricated Steel Tower
Bolted Steel Tank
Sectionalized, Knocked Down Tower
Bolted Connections
Field Erections

Custodian:

Navy - YD

Preparing Activity:

Navy - YD

(Project 5445-N023)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-82096B(YD)	TOWER, PREFABRICATED, STEEL: FOR 4200 GALLON WATER TANK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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